

Skill Check 3: Building Better Code

Exam Agreement

This Skill Check is an individual assessment and you should not receive or offer help on it from any other human.

You may use any **physical** resource to complete the work. This includes:

- Any notes, code, slides, papers, or previous feedback from the instructor as long as they are on paper.
- Any books that you have with you.
- Any scholarly works such as papers that you have with you.

You may NOT use:

- Help from any other student or person. **This is an individual assessment.**
- Any digital resource that does not exist as a physical copy present in class.
- The use of generative artificial intelligence (e.g., ChatGPT).
- Asking for help on specific questions. (It is fine to ask for clarification!)
- Help from homework websites such as Course Hero or Chegg.

By signing, you agree that you have neither given nor received unauthorized aid on this examination.

Printed Name: _____

Chapman ID: _____

Signed: _____

Date: _____

Include this signed page as the first page of your submitted work.

Skill Check Instructions

You must answer every question on the exam to the best of your ability. Include all appropriate code, syntax, and functions that would lead to the code to run successfully. For each two questions correct, you will receive one course point for a total of two course points.

Question 1: FUNCTIONS

Write a function that returns the average of the biggest and smallest numbers in a numeric vector. Include an example function call with a vector of length 4 to demonstrate the function in operation.

[illegible]

Question 2: LOOPS

Create a loop that prints out all even numbers up to and including 100.

[illegible]

Question 3: CONDITIONALS

Generate 10 random numbers between 12 and 25 using `sample()` and use a conditional to create a vector named `is_even` that holds “Yes” where the randomly generated numbers are even and “No” where they are odd.

[illegible]

Question 4: COMBINATIONS

Use a conditional statement to color the points on a plot of Displacement (disp) versus Horsepower (hp) in the mtcars data set based on whether they have more than 6 cylinders (in the cyl column) or fewer than 6 cylinders.

[illegible]

[illegible]